

Final Project - Unsupervised Learning

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Abstract

In this project we were given 3 different datasets: Static, Dynamic and Graph based. On each one we performed clustering analysis using different clustering methods and explained which one is the best one. We associated the clusters with the ground truth, and explained what external variable is best associated with clusters, and found which one of the clustering methods best associates with the outside variables. After that, we found anomalies in the data using Anomaly detection tools, tested whether anomalies are associated with any of the external variables and finally checked if anomaly detection improves the clustering. Lastly we proposed a visualization that best characterizes the clusters associated with the external variables. We concluded that Minibatch KMeans is overall the best algorithm for the Static dataset, and KMeans was best for both the Graph based dataset and the Dynamic dataset. As for the external variables, both the Dynamic dataset and Graph based dataset had only one external variable which were Genre and Gender accordingly, the Dynamic dataset had two and the best one was the Gas type variable. The code is available at https://github.com/ExtraFlash/Or_Ohad_Unsupervised_Static_database, https://github.com/ExtraFlash/Or_Ohad_Unsupervised_Graph, https://github.com/ExtraFlash/Or_Ohad_Unsupervised_Dynamic.

1 Introduction

Unsupervised learning refers to the use of artificial intelligence algorithms to identify patterns in datasets containing data points that are neither classified nor labeled. The algorithms are thus allowed to classify, label and group the data points contained within the datasets without having any external guidance in performing that task [1]. In this project we use three different aspects of this field: Clustering, Dimensionality reduction and Anomaly detection tools. Clustering is the task of dividing the population or data points into a number of groups such that data points in the same groups are more similar to other data points in the same group and dissimilar to the data points in other groups. It is basically a collection of objects on the basis of similarity and dissimilarity between them [2]. Dimensionality reduction is the process of reducing the number of dimensions in a dataset while retaining as much information as possible. This can be done for a variety of reasons, such as to reduce the complexity of a model, to improve the performance of a learning algorithm, or to make it easier to visualize the data [3]. Anomaly detection is generally understood to be the identification of rare items, events or observations which deviate significantly from the majority of the data and do not conform to a well defined notion of normal behaviour [4]. In this project, for each dataset we use a dimensionality reduction algorithm called IPKA to reduce the number of features [5]. We further apply 4 different clustering methods: GMM, KMeans, Birch and Minibatch KMeans and apply statistical tests to determine the best clustering method according to the Silhouette score and see how clusters relate to the external variables according to the Adjusted MI score. We also used anomaly Detection tools to determine whether anomalies are associated with any of the external variables and see if they improve our clustering results.

2 Methods

2.1 FMA: A Dataset For Music Analysis

This dataset was collected from a research in 2017 of International Society for Music Information Retrieval Conference (ISMIR) [6], there are 106,577 songs, each one with 140 numerical features and a label of genre (16 different genres). Some of the rows contained null value at the genre label, so the way we handled this was to remove them and stay with 49,598 songs. Afterwards we shuffled the rows and took the first 32,238 songs (65%) for a train set and the rest 17,360 songs (35%) for a test set.

2.2 Deezer Ego Nets

The ego-nets of Eastern European users collected from the music streaming service Deezer in February 2020. Nodes are users and edges are mutual follower relationships [7]. This dataset contains 9,629 ego-nets of users, each one with a gender label of 0 or 1. For each graph, we calculated the eigenvalues of the normalized Laplacian, sorted it in a reverse order, took the first 50 elements and normalized the vector. Additionally, we concatenated the average clustering coefficient of the graph to

our vector and finally added the gender label. This way we translated the set of graphs to a set of vectors in Euclidean space, each containing 51 numerical features and the genre label. We then shuffled the rows and splitted the data, 65% went to a train set and the rest 35% to a test set.

2.3 Gas Sensor Array Drift Dataset at Different Concentrations Data Set

This data set contains 13,910 measurements from 16 chemical sensors exposed to 6 gases at different concentration levels. This dataset is providing the information about the concentration level at which the sensors were exposed for each measurement [8]. The dataset is organized into ten batches, each containing the number of measurements per class and month. First we combined all the batches into a single csv file where each row represents a measurement and containing 128 numerical features and 2 labels: gas type (6 different types) and concentration level (59 different concentration levels). We then shuffled the rows and splitted the data, 65% went to a train set and the rest 35% to a test set.

2.4 Statistical tests

- To examine if there is a difference between all the clustering methods performances, we used a one-way ANOVA and Kruskal-Wallis.
- To determine which one of the 2 best clustering methods is the better one, we used a paired t-test on two related samples of scores and Mann-Whitney U test.

2.5 Finding the optimal number of clusters and dimensions

In order to get the optimal number of clusters for each clustering method separately, we ran all the four clustering methods each time with a different number of clusters, calculated the silhouette score and after that, we picked the optimal number of clusters for each method using the Elbow method. In order to get the optimal number of dimensions for each clustering method separately, we ran IPCA each time with a different number of dimensions, then ran all the clustering methods on the low-dimensional data (each one with it's optimal number of clusters) and calculated the silhouette scores. Finally, for each clustering method we picked the number of dimensions used by IPCA that gave us the highest silhouette score.

2.6 Anomaly Detection

We used two different anomaly detection methods:

- **Isolation Forest**
- **One-class SVM** We set an upper bound of 0.01 on the fraction of training errors and a lower bound of the fraction of support vectors

3 Results

3.1 Finding the optimal number of clusters and dimensions

For finding the optimal number of clusters, we ran all the clustering methods, each time with a different number of clusters.

For the Static dataset: [2, 3, 4, ..., 16].

For the Graph based dataset: [2, 3, 4, ..., 10]

For the Dynamic dataset: [5, 8, 11, ..., 59]

After using the Elbow method, we got the optimal number of clusters for each clustering method (see Table 1, Table 2, Table 3). Afterwards, we performed the IPCA algorithm with each clustering method with the optimal number of clusters we recieved earlier with different number of dimensions and found the best number of dimensions based on the silhouette score (see Table 1, Table 2, Table 3).

Table 1: Static dataset

	GMM	KMeans	Birch	Minibatch KMeans
Number of clusters	4	7	5	4
Number of dimensions	5	100	5	5

Table 2: Graph based dataset

	GMM	KMeans	Birch	Minibatch KMeans
Number of clusters	5	6	7	7
Number of dimensions	5	5	5	25

Table 3: Dynamic dataset

	GMM	KMeans	Birch	Minibatch KMeans
Number of clusters	29	14	32	20
Number of dimensions	70	50	20	70

3.2 Finding the best clustering algorithm

Using the test data, we divided it into disjoint groups (30 groups for the Static dataset, 40 groups for the Graph based dataset and 60 groups for the Dynamic dataset) and we ran each clustering method on every group with optimal number of clusters, after applying the IPCA algorithm with optimal number of dimensions and calculated the silhouette scores. For each clustering method we have a vector of silhouette scores, we performed one-way ANOVA test between the vectors of each clustering method to showcase that one of them is better than the others. For the static dataset, our result is significant with $p < 1.59e - 26$. For the graph based dataset, our result is significant with $p < 1.35e - 91$. For the dynamic dataset, our result is significant with $p < 7.22e - 3$. Afterwards, we performed the paired t-test between the vectors of results of the two best clustering methods to determine the best clustering method. For the static dataset, we got that Minibatch KMeans is better than Birch with a significant result of $p < 3.73e - 14$. For the graph based dataset, we got that KMeans is better than Minibatch KMeans with a significant result of $p < 1.43e - 26$. For the dynamic dataset, we got that KMeans is better than Minibatch KMeans with a significant result of $p < 3.06e - 16$. Lastly, we proposed a weighted score of the silhouette and the MI of the clusters with the best associated external variable (see Association between clusters and external variables) in order to determine the best clustering algorithm according to both the silhouette and MI.

For the static dataset, we divided the test data into 30 groups, for each clustering method we calculated $weighted = 0.7 * silhouette + 0.3 * MI$ and afterwards performed one-way ANOVA test between the results and got a significant result of $p < 1.37e - 23$. Following that, we performed the paired t-test between the vectors of results of the two best clustering methods and got that Minibatch KMeans is better than KMeans with a significant result of $p < 1.53e - 11$, therefore Minibatch KMeans is overall the best algorithm in the static dataset.

For the graph based dataset, we divided the test data into 40 groups, for each clustering method we calculated the silhouette score and the adjusted MI with the gender label and defined $weighted = 0.7 * silhouette + 0.3 * MI$ and performed one-way ANOVA test with a significant result of $p < 8.14e - 92$. Following that, we performed the paired t-test between the vectors of results of the two best clustering methods and got that Minibatch KMeans is better than KMeans with a significant result of $p < 6.63e - 26$, therefore Minibatch KMeans is overall the best algorithm in the graph based dataset.

For the dynamic dataset, we divided the test data into 60 groups, for each clustering method we calculated the silhouette score and the adjusted MI with the gas type label and defined $weighted = 0.5 * silhouette + 0.5 * MI$ and afterwards performed one-way ANOVA test with a significant result of $p < 3.34e - 3$. Following that, we performed the paired t-test between the vectors of results of the two best clustering methods and got that Minibatch KMeans is better than KMeans with a significant result of $p < 4.18e - 10$, therefore Minibatch KMeans is overall the best algorithm in the dynamic dataset.

3.3 Association between clusters and external variables

Using the test data, we divided it into disjoint groups and we ran each clustering method on every group with optimal number of clusters, after applying the IPCA algorithm with optimal number of dimensions and calculated the adjusted MI with the external variables. Afterwards, for each external variable, we ran a statistical test each between the vectors of results of each clustering method to showcase that one of them is better than the others. For the static dataset, we divided the test data into 30 groups and for each clustering method calculated the adjusted MI with the genre label and afterwards performed one-way ANOVA test with a significant result of $p < 1.71e - 2$. Following that, we performed the paired t-test between the two best clustering methods and got that KMeans is better than Minibatch KMeans with a significant result of $p < 5.83e - 3$, therefore KMeans best associates with the genre label. For the graph based dataset, we divided the test data into 40 groups and for each clustering method calculated the adjusted MI with the gender label and afterwards performed Kruskal-Wallis test between the results and showcased that one of them is better than the others with a significant result of $p < 3.92e - 3$. Following that, we performed the Mann-Whitney U test between the vectors of results of the two best clustering methods

and got that Minibatch KMeans is better than GMM with a significant result of $p < 6.22e - 4$, therefore Minibatch KMeans best associates with the gender label. For the dynamic dataset, first we divided the test data into 9 groups and for each clustering method calculated the adjusted MI with the gas type label and afterwards performed Kruskal-Wallis test between the results and showcased that one of them is better than the others with a significant result of $p < 1.16e - 3$. Following that, we performed the Mann-Whitney U test between the vectors of results of the two best clustering methods and got that GMM is better than Minibatch KMeans with a significant result of $p < 6.66e - 3$, therefore GMM best associates with the gas type label. Secondly, we divided the test data into 50 groups and for each clustering method calculated the adjusted MI with the concentration level label and afterwards performed Kruskal-Wallis test between the results and showcased that one of them is better than the others with a significant result of $p < 3.10e - 3$. Following that, we performed the paired t-test between the vectors of results of the two best clustering methods and got that Birch is better than Minibatch KMeans with a significant result of $p < 4.20e - 6$, therefore Birch best associates with the concentration level label. Lastly, we took the results of adjusted MI between the GMM clusters and the gas label and the results of adjusted MI between the Birch clusters and the concentration level label and performed a paired t-test. We got that GMM is better than Birch with a significant result of $p < 7.86e - 7$, therefore the gas type label is best associated with clusters and GMM best associates with the outside variables.

3.4 Anomaly Detection

In each dataset, we performed two anomaly detection methods on the test set in order to find anomalies in the data, tested whether anomalies are associated with any of the external variables and checked if anomaly detection improved the clustering.

- Isolation Forest** For the static dataset, after performing the Isolation Forest method, we got that 700 songs out of 17,360 were anomalies. Following that, we computed the adjusted MI between the anomalies and the genre external variable and received a score of $1.38e - 2$, and we performed IPCA and Minibatch KMeans (the overall best algorithm, with the optimal amount of clusters and dimensions) on the test data excluding the anomalies and received a silhouette score of $5.57e - 1$ which is the same score that we got without removing anomalies, therefore we can conclude that the anomalies aren't associated with the genre label and that it did not improve the clustering results.
 For the graph based dataset, after performing the Isolation Forest method, we got that 632 graphs out of 3,370 were anomalies. Following that, we computed the adjusted MI between the anomalies and the gender external variable and received a score of $1.04e - 3$, and we performed IPCA and KMeans (the overall best algorithm, with the optimal amount of clusters and dimensions) on the test data excluding the anomalies and received a silhouette score of $5.48e - 1$ which is less than the score that we got without removing anomalies ($5.52e - 1$), therefore we can conclude that the anomalies aren't associated with the gender label and that it did not improve the clustering results.
 For the dynamic based dataset, after performing the Isolation Forest method, we got that 665 measurements out of 5,564 were anomalies. Following that, we computed both the adjusted MI between the anomalies and the gas type external variable and received a score of $2.44e - 2$ and the adjusted MI between the anomalies and the concentration level external with a score of $6.94e - 2$, and we performed IPCA and KMeans (the overall best algorithm, with the optimal amount of clusters and dimensions) on the test data excluding the anomalies and received a silhouette score of $3.48e - 1$ which is less than the score that we got without removing anomalies ($3.49e - 1$), therefore we can conclude that the anomalies aren't associated with any of the external variables and that it did not improve the clustering results.
- One-class SVM** For the static dataset, after performing the One-class SVM method, we got that 173 songs out of 17,360 were anomalies. Following that, we computed the adjusted MI between the anomalies and the genre external variable and received a score of $2.02e - 5$, and we performed IPCA and Minibatch KMeans (the overall best algorithm, with the optimal amount of clusters and dimensions) on the test data excluding the anomalies and received a silhouette score of $5.57e - 1$ which is the same score that we got without removing anomalies, therefore we can conclude that the anomalies aren't associated with the genre label and that it did not improve the clustering results.
 For the graph based dataset, after performing the One-class SVM method, we got that 34 graphs out of 3,370 were anomalies. Following that, we computed the adjusted MI between the anomalies and the gender external variable and received a score of $9.40e - 4$, and we performed IPCA and KMeans (the overall best algorithm, with the optimal amount of clusters and dimensions) on the test data excluding the anomalies and received a silhouette score of $5.47e - 1$ which is less than the score that we got without removing anomalies ($5.52e - 1$), therefore we can conclude that the anomalies aren't associated with the gender label and that it did not improve the clustering results.
 For the dynamic based dataset, after performing the One-class SVM method, we got that 54 measurements out of 5,564 were anomalies. Following that, we computed both the adjusted MI between the anomalies and the gas type external variable and received a score of $6.55e - 2$ and the adjusted MI between the anomalies and the concentration level external with a score of $6.62e - 3$, and we performed IPCA and KMeans (the overall best algorithm, with the optimal amount of clusters and dimensions) on the test data excluding the anomalies and received a silhouette score of $3.45e - 1$ which is less than the score that we got without removing anomalies ($3.49e - 1$), therefore we can conclude that the anomalies aren't associated with any of the external variables and that it did not improve the clustering results.

Figure 1: Static dataset

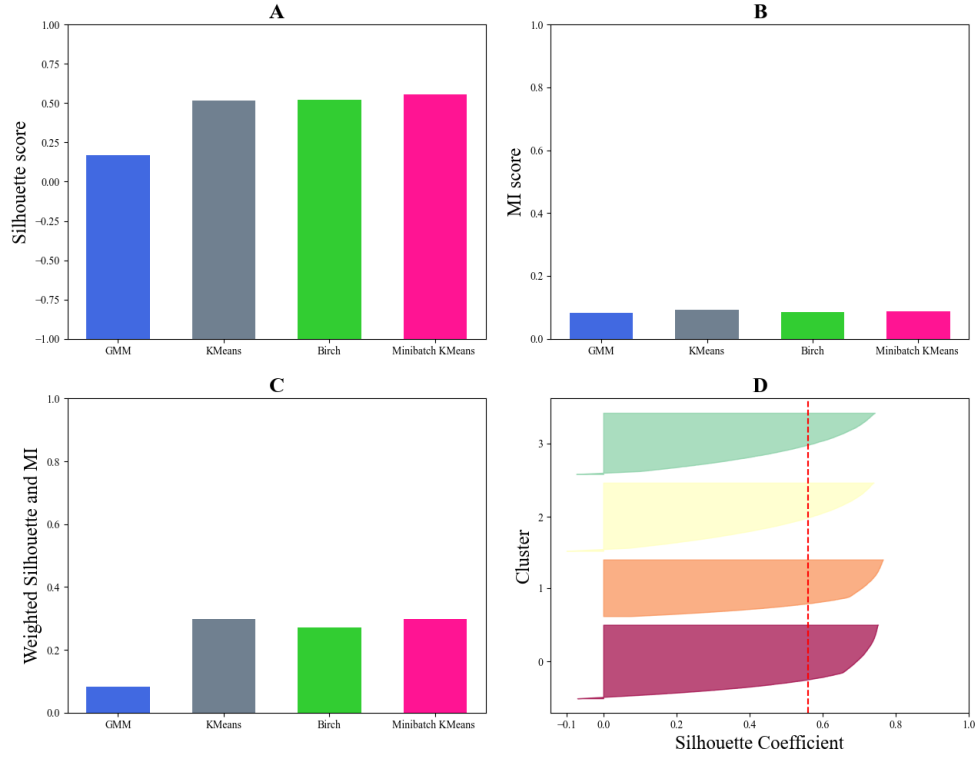


Figure 1: **A.** Clustering results. **B.** adjusted MI between clustering methods and genres. **C.** Weighted score between clustering methods and genres. **D.** Silhouette visualization of the best algorithm, Minibatch KMeans.

Figure 2: Graph based dataset

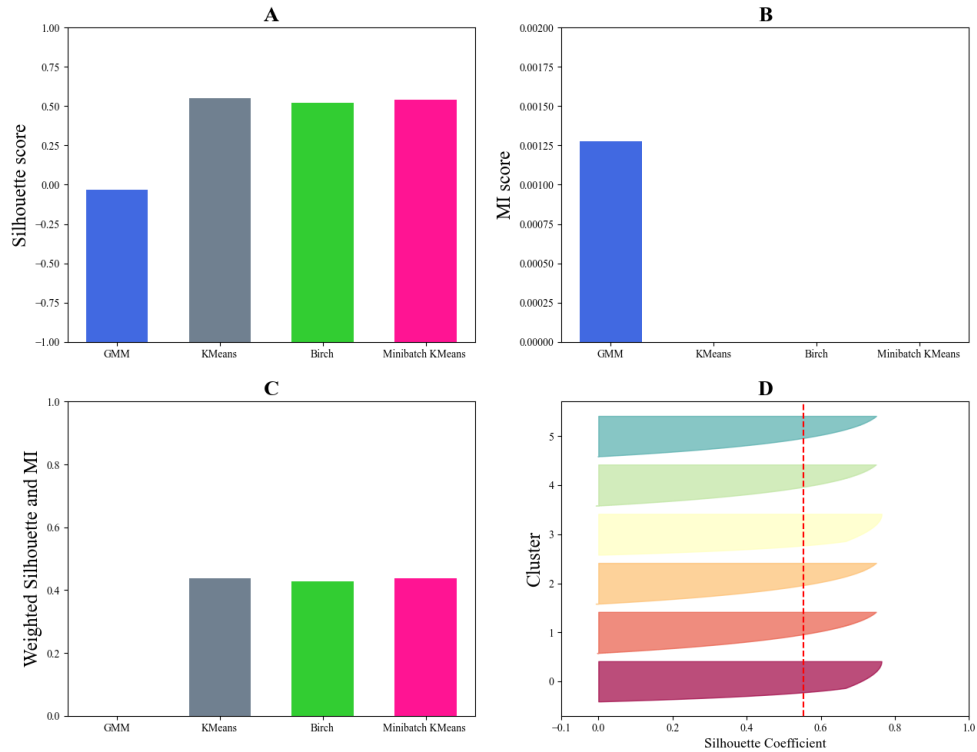


Figure 2: **A.** Clustering results. **B.** adjusted MI between clustering methods and genders. **C.** Weighted score between clustering methods and genders. **D.** Silhouette visualization of the best algorithm, KMeans.

Figure 3: Dynamic dataset

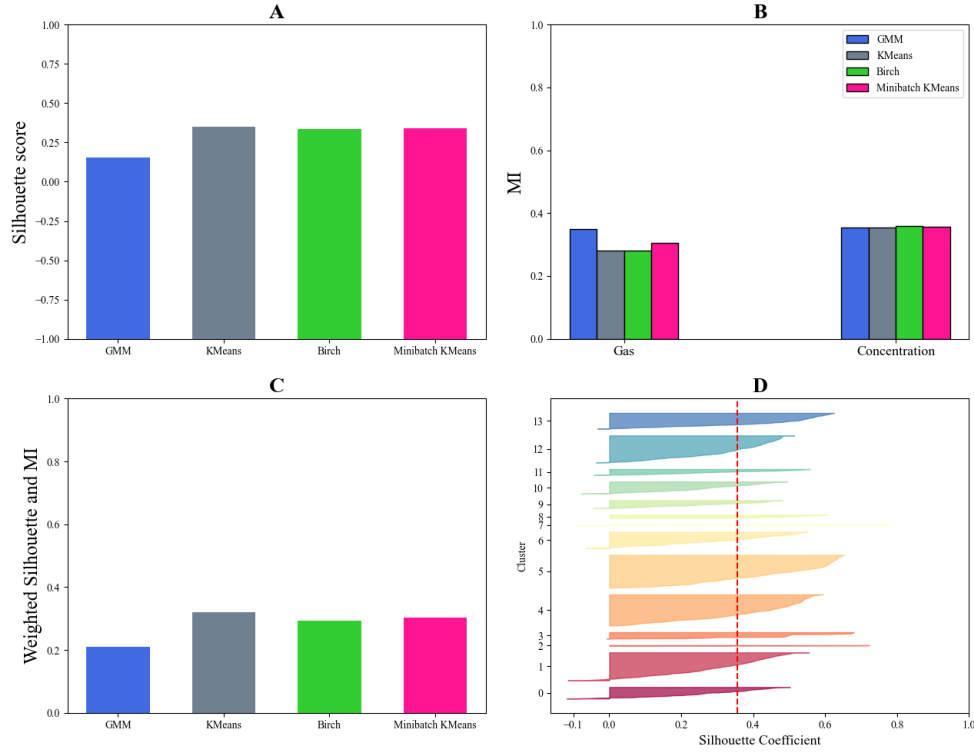


Figure 3: **A.** Clustering results. **B.** adjusted MI between clustering methods and gas types. **C.** Weighted score between clustering methods and gas types. **D.** Silhouette visualization of the best algorithm, KMeans.

3.5 Visualization

We used the IPCA algorithm on the test data and performed clustering with the best overall clustering method (with optimal number of dimensions and with the same number of clusters as the amount of categories in the most associated external variable), We reduced the dimensions of the data into 2 dimensions and visualized it in order to show the correlation between the best associated external variable and the best associated clustering method. In the plot, big dots with bright colors represent association to clusters and small dots with dark colors represent association to different categories of the external variable. For the static dataset we set the amount of clusters to 16, as the number of genres and used Minibatch KMeans for the clustering (Figure 4).

For the graph based dataset we set the amount of clusters to 2, as the number of genres and used KMeans for the clustering (Figure 5).

For the dynamic based dataset we set the amount of clusters to 6, as the number of gas types and used KMeans for the clustering (Figure 6).

Figure 4: Static dataset

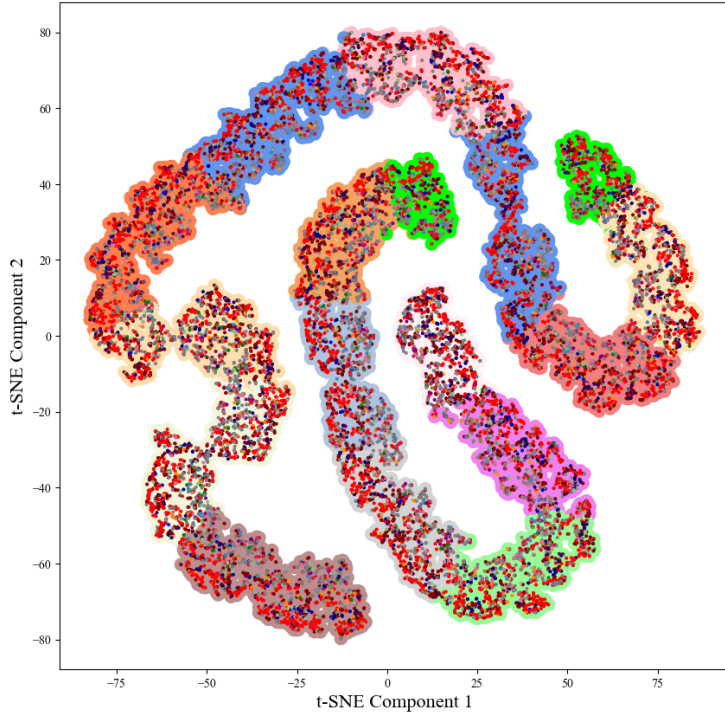


Figure 5: Graph based dataset

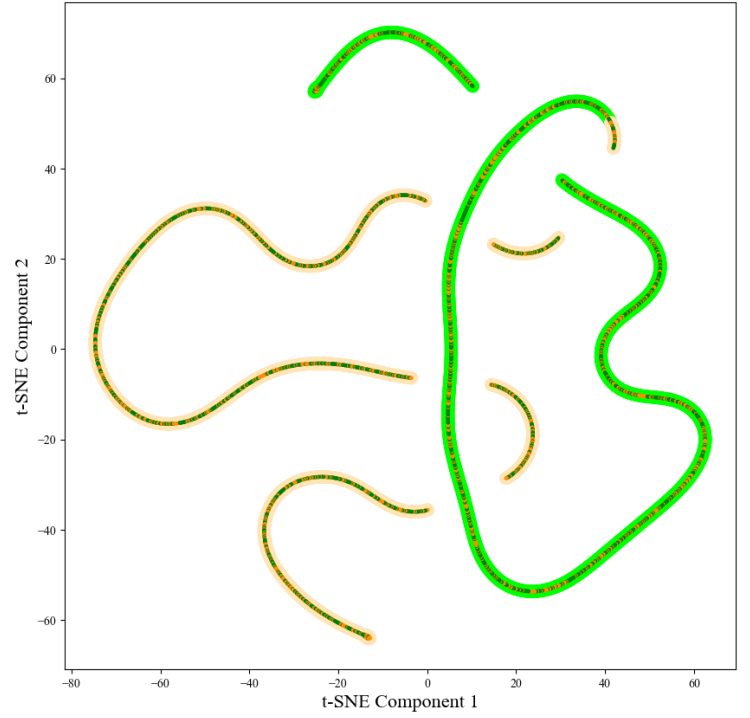
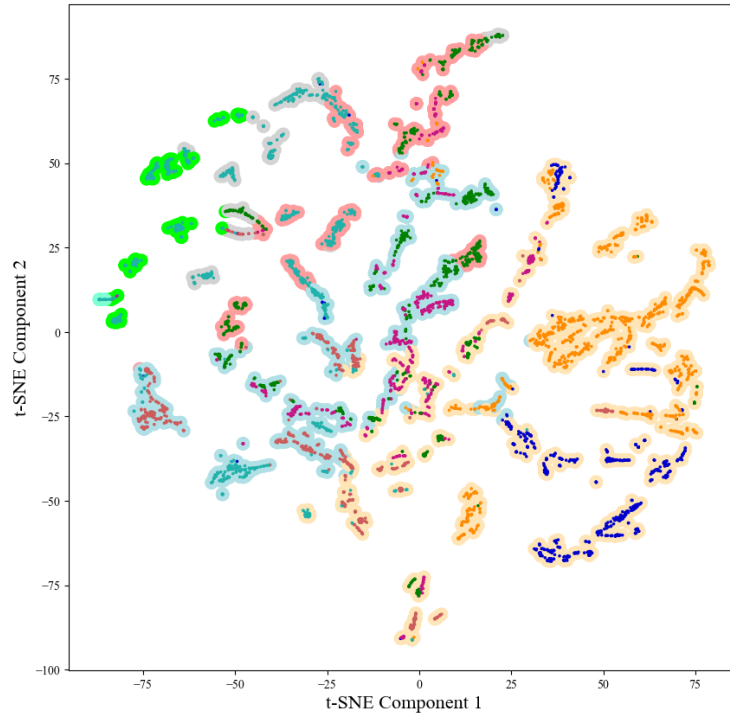


Figure 6: Dynamic dataset



4 Discussion

In this project we implemented unsupervised learning tools such as clustering, dimensionality reduction and anomaly detection. For each dataset we ran different clustering methods, for each method we found the optimal number of clusters together with optimal number of dimensions after performing the IPCA algorithm, according to the silhouette score. It seems that the silhouette score is biased towards small amount of clusters, therefore we used the Elbow method in order to get the optimal

number of clusters. For the static dataset we got that Minibatch KMeans performs the best according to the silhouette score, with 4 clusters and 5 dimensions. For the graph based dataset we got that KMeans performs the best according to silhouette, with 5 clusters and 5 dimensions. For the dynamic dataset we got that KMeans performs the best according to silhouette, with 14 clusters and 50 dimensions. We then defined a weighted score for each dataset that takes into account both silhouette score and MI with the most associated external variable. The reason we defined it is that although we got decent results with silhouette, in the static and graph based datasets we got low results of MI and we wanted a measure that takes into account both the quality of the clustering and association to external variables which seems more reliable. This implies that a getting a high silhouette score does not imply association with outside variables. We noticed that Minibatch KMeans and KMeans gave the best results both with the silhouette and the weighted score, and the reason is that the MI scores that we got were usually lower than the silhouette scores. We also checked what external variable is best associated with clusters, and what clustering method best associates with the outside variables with using the adjusted MI score. For the static dataset, we got that KMeans best associates with genres. For the graph based dataset, we got that Minibatch KMeans best associates with genders. For the dynamic dataset, we got that GMM and gas type external variable best associates with each other. Later we used anomaly detection tools such as Isolation Forest and One-class SVM on each of the datasets. We found that the anomalies have low correlation with the external variables, moreover, removing the anomalies didn't give us better clustering results. Therefore we concluded that using anomaly detection tools will not always improve our measurements. Lastly, we proposed a visualization of the data that indicates whether there is a correlation with clusters and external variables (In our case we used the best associated external variable for each dataset). We found out that when there is a high silhouette score, the clusters seems more separated and form bigger shapes. When we get a low MI score, we see that the little dark dots seems to be mixed in the clusters, and when the MI score is high, we can see that there is a similarity between between the little dots (The external variable) and the bigger dots (the clusters).

References

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